

Ben Zhou

ben.zhou@uwaterloo.ca | bzhou.ca | github.com/benz206 | linkedin.com/in/ben-zhou06/

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, HTML, CSS, C++, C, Python, Java, SQL, GraphQL, Rust

Tech: Next.js, React, Vite, Flask, Express, Mongo, Redis, Tailwind, SQLite, Postgres, Prisma, Drizzle

Development: Git, GCP, AWS, Posthog, Oracle, Cloudflare, Linux, Firebase, Vercel, Vitest, Jest, ESLint, Docker

EXPERIENCE

Software Engineering Intern

Sep 2025 – Present

Grand Charter

New York City, New York

- Built a production email integration for Gmail & Outlook, with context-aware drafting, attachment streaming, file/folder navigation and multi-account support, cutting document ingestion by over **78%**.
- Provided **FDE** services to deploy tailored solutions, augmenting existing products with varying **tech stacks**.
- Built **E2E intake pipelines** for multiple case types with built-in integrations and tooling.
- Created new landing page templates and updated **CMS**, increasing impressions/clicks by **20-30%** monthly. x
- Developed an **linting/static-analysis tool** to prevent unsafe db action usage, improving CI and code safety.
- Delivered **20+ external-facing APIs** and authored developer documentation to support third-party and internal integrations at scale.

Software Engineering Intern

Jan 2025 – Apr 2025

Fuego.io

San Francisco, California

- Optimized core AI generation features by developing custom architecture, accelerating response times to be **17.7x faster (15.4s → 0.87s avg)**, reducing token usage and slashing costs by similar margins.
- Created a dashboard to log, render, sort and debug **1,000+** AI conversations, adding tagging and statistic cards.
- Wrote **120 Jest/Vitest tests (100% branch cov.)** for timezone & metafield logic, eliminating recurring bugs.
- Implemented ESLint, and resolved **200+** production warnings and errors to enhance code quality.
- Built **audience segmentation** features, enabling targeted notifications for brands reaching **millions** of users.
- Wrote currency conversion logic with constant metadata updates via a cron scheduling script using **GraphQL**.
- Updated UI components for analytics, allowing **advanced date comparisons** with shareable URLs.

Software Developer Co-op Student

Feb 2024 – Jul 2024

SAP Toronto

Toronto, Ontario

- Created a worker to handle and sanitize **GPT-4o** requests using **TypeScript** reducing request errors by **23%**.
- Wrote comprehensive unit tests using Vitest to achieve **100% coverage** of request worker.
- Wrote scripts in **Python** and **Node** to clean **248,000** records, updated schema to accelerate processing by **26%**.

FullStack Developer Lead

Nov 2023 – May 2024

EurekaHacks 2024

Oakville, Ontario

- Improved page load times by **160%**, and reduced LCP, leading to **3,800+ impressions** and **1,100+ clicks**.
- Led a team to build a modern, responsive website using **Next.js**, secured a max event capacity of **300+ sign-ups**.

PROJECTS

GooseontheLoose - HackTheNorth '24 Winner | [GitHub](#) | *Next, Mongo, TypeScript*

- Designed an API in **TypeScript** to classify and recognize Canadian geese with over **95%** accuracy.
- Integrated Google Maps API with **MongoDB** to store geolocal data for accurate real-time tracking.

bTagScript Lang | [GitHub](#) | [Project](#) | *Flask, Python, HTML/CSS, JavaScript, MySQL*

- Designed a **RESTful Flask API** to run sandboxed programs, logging runtime data through **MySQL**.
- Processed **15,270+ programs**, and recently migrated to a new host to ensure **99.98%** uptime for all users.

Tag Vault | [GitHub](#) | [Project](#) | *TypeScript, Mongo, Next.js, Tailwind*

- Deployed a scraper to monitor over **2 million scripts** and archive an additional **35,900+** user-uploaded scripts.
- Built a search, filter, and favorites module with OAuth SSO, enabling **100 % authenticated access**.
- Implemented a time-based cache system to lower API requests by over **90%** and improving fetch time by **3s**.

Spotify Macrokeyboard | [GitHub](#) | [Project](#) | *C++, TypeScript*

- Implemented an API in **TypeScript** to handle Spotify states and retrieve currently playing music.
- Rewrote a library in **C++** to accelerate API requests & processing by over **750% (15s → 2s)**.

EDUCATION

University of Waterloo

Sep 2024 – Apr 2029

Candidate for BAsC in Computer Engineering, Honours, In Excellent Standing with Distinction

Waterloo, Ontario